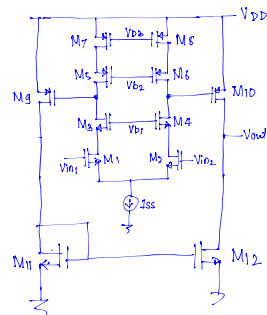
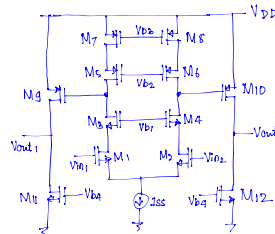


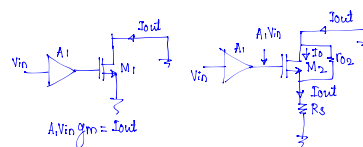
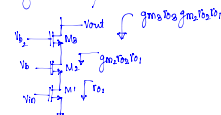
```

graph LR
    Vin[V_in] --> A1[High gain Amplifier]
    A1 --> Vmid[V_mid]
    Vmid --> A2[High stable Amplifier]
    A2 --> Vout[V_out]
  
```



boosting :-

$$A_v = G_m \times R_{out}$$
 Needs to be improved
 very hard to improve



$$\frac{I_{out}}{V_{in}} = A_{1gm}$$

$$I_o = (A_1 V_{in} - I_{out} R_s) g_{m2}$$

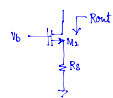
$$I_{out} = I_o \times \frac{R_{o2}}{R_{o2} + R_s}$$

$$= (A_1 V_{in} - I_{out} R_s) g_{m2} \frac{R_{o2}}{R_{o2} + R_s}$$

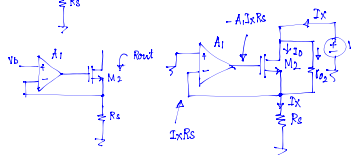
$$T_{out} \left[1 + \frac{g_{m2} R_{o2} R_s}{R_{o2} + R_s} \right] = A_{1, Vin} g_{m2} \frac{R_{o2}}{R_{o2} + R_s}$$

$$\frac{I_{out}}{V_{in}} = A_v g_{m2} \frac{r_{o2}}{r_{o2} + R_S} \times \frac{r_{o2} + R_S}{r_{o2} + R_S + g_{m2} r_{o2} R_S}$$

$$= \frac{A_1 g_{m2} r_{o2}}{r_{o2} + R_S + g_{m2} r_{o2} R_S} \quad \text{No improvement}$$



$$R_{out} = R_s + r_o + g_m R_s r_o$$



$$I_o = g_{m2} \int -A_v I_x R_s - I_x R_s$$

$$= -g_{m2} R_S I_X [1 + A_v]$$

$$I_x = I_0 + \frac{V_x - I_x R_S}{R_{02}}$$

$$I_x = -g_{m2} R_S I_x [1 + A_1] + \frac{V_x - I_x R_S}{R_{02}}$$

$$\frac{V_x}{I_x} = R_s + r_{o2} + g_{m2}(1+A_1)R_s r_{o1}$$